The Kernel Trick for Nonlinear Factor Modeling

Varlam Kutateladze[†]

[†]University of California, Riverside

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Factor Modeling

Idea: Variation in $\underset{N\times 1}{X_t}$ is mainly due to $r \ll N$ latent variables.

$$\underset{N\times 1}{X_t} = \underset{N\times r}{\Lambda}\underset{r\times 1}{F_t} + \underset{N\times 1}{e_t}$$

- Psychology (Thompson 1938) · Statistics (Lawley and Maxwell 1962) · Biology (Hirzel et al. 2002) · Economics & Finance (Chamberlain and Rothschild 1983, Fama and French 1992, Stock and Watson 2002a, Fan et al. 2013)

For Forecasting:

- Handles "Big Data" / Dimensionality reduction
- Consistency (Connor and Korajczyk 1986, Forni et al. 2000, Stock and Watson 2002a, Bai and Ng 2006)
- Strong performance (Stock and Watson 2002b, Kim and Swanson 2018, Coulombe et al. 2019)
- Readily estimable
- DSGE stylized fact (Sargent and Sims 1977)

Motivation

"Further forecast improvements will need to come from models with nonlinearities and/or time variation."

Stock and Watson 2012

"Nonlinear factor-augmented regression should be considered."

Cheng and Hansen 2015

- ✓ Nonlinear models often dominate their linear counterparts: Teräsvirta et al. 1994, Giovannetti 2013, Kim and Swanson 2014, Coulombe et al. 2019.
- ✓ Some attempts of nonlinear modeling: Yalcin and Amemiya 2001, Bai and Ng 2008, Exterkate et al. 2016

This study: Nonlinear factor modeling via the kernel method.

- ✓ General approach that nests the linear factor model
- ✓ Factor estimator is consistent
- ✓ Improved forecasting of the key macroeconomic variables

PCA

Linear factor model:

$$\underset{T\times N}{X} = \underset{T\times r}{F}\underset{r\times N}{\Lambda'} + \underset{T\times N}{e}, \tag{1}$$

where $X = [X_1, \ldots, X_T]'$ and $F = [F_1, \ldots, F_T]'$.

Estimation problem:

$$\begin{gathered} \arg\min_{F,\Lambda} \|X - F\Lambda'\|_F^2 \\ N^{-1}\Lambda'\Lambda = I_r, \quad F'F \text{ diagonal}, \end{gathered} \tag{2}$$

Solution:

$\widehat{F}_{PCA} = PC_r(X) = Xeig_r(X'X)$.

- Bai and Ng 2008 : $\widehat{F}_{SPCA} = PC_r([X,\ X^2])$.
- Nonlinear estimates?

Kernel Method

Idea: Implicit nonlinearization of inputs $\varphi(\cdot) : \mathcal{X} \to \mathcal{F}$ $(\mathbb{R}^N \to \mathbb{R}^M)$.

How: Substitute $\langle \mathrm{x}_i, \mathrm{x}_j \rangle$ with $\langle \varphi(\mathrm{x}_i), \varphi(\mathrm{x}_j) \rangle = k(\mathrm{x}_i, \mathrm{x}_j)$.

- *Example:* RBF kernel $k(\mathrm{x_i}, \mathrm{x_j}) = e^{-\gamma ||\mathrm{x_i} - \mathrm{x_j}||^2}$
- Calculate Gram matrix K: $K_{ij} = k(\mathrm{x}_i, \mathrm{x}_j) = \varphi(\mathrm{x}_i)'\varphi(\mathrm{x}_j)$
- *Example:* Ridge prediction for x_*:
  - Standard version: $\widehat{Y} = x_*(X'X + \lambda I_N)^{-1}X'Y$
  - Alternative version: $\widehat{Y} = \mathrm{x}_*{}'X'(XX' + \lambda I_T)^{-1}Y$
  - Kernel version: $\widetilde{Y} = k_*'(K + \lambda I_T)^{-1}Y$

Mercer's condition: $\int\int f(\mathrm{x}_i)k(\mathrm{x}_i, \mathrm{x}_j)f(\mathrm{x_j})d\mathrm{x}_i d\mathrm{x_j} \geq 0$, for any square-integrable function $f(\cdot)$.

Kernel method

Example: Polynomial kernel of degree 2, $k(\mathrm{x_i}, \mathrm{x_j}) = (\mathrm{x_i}'\mathrm{x_j})^2$

Nonlinearity is $\varphi(\mathrm{x_i}) = (x_{i1}^2,\ \sqrt{2}x_{i1}x_{i2},\ x_{i2}^2)'$, since $\varphi(\mathrm{x_i})'\varphi(\mathrm{x_j}) = (\mathrm{x_i}'\mathrm{x_j})^2$

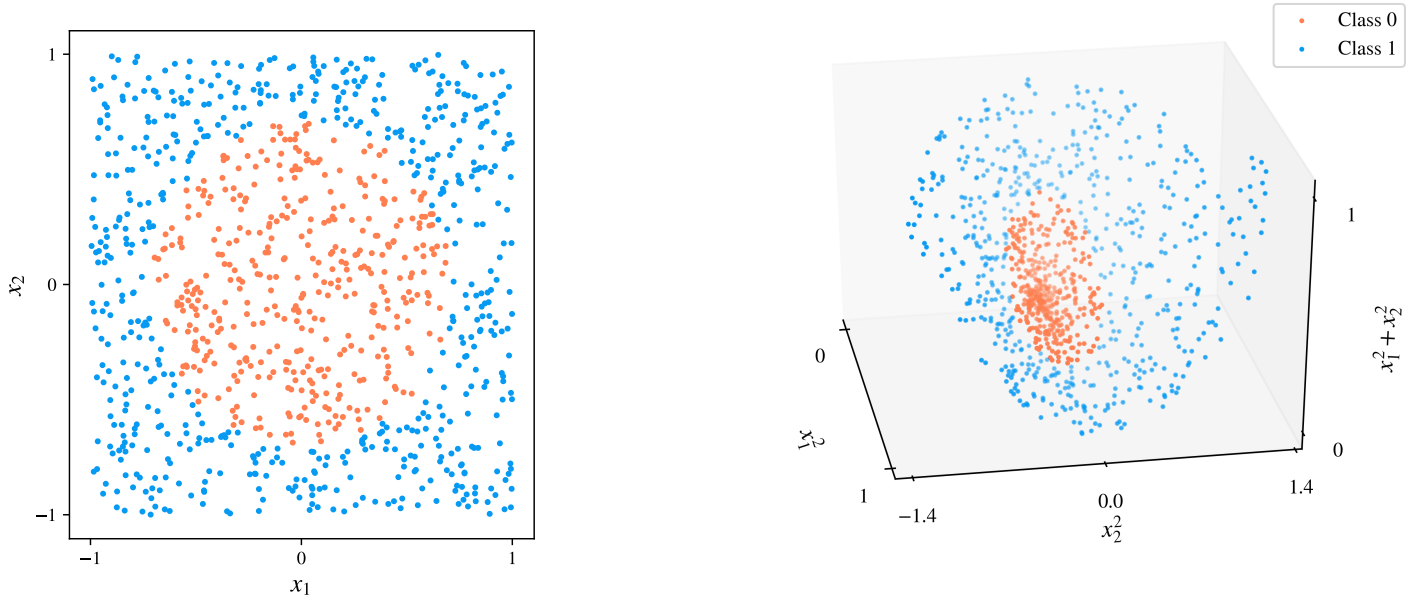

Figure 1: Left: observations from two classes in the original space, not linearly separable. Right: observations in the feature space, linearly separable.

Nonlinearities

Wide range of flexible nonlinearities, controlled by a hyperparameter.

Example, RBF kernel:

$$k(\mathrm{x_i}, \mathrm{x_j}) = e^{-\gamma \|\mathrm{x_i} - \mathrm{x_j}\|_2^2}$$
$$= \langle \varphi(\mathrm{x_i}), \varphi(\mathrm{x_j}) \rangle_{\mathcal{F}}$$

$$\varphi(x) =$$

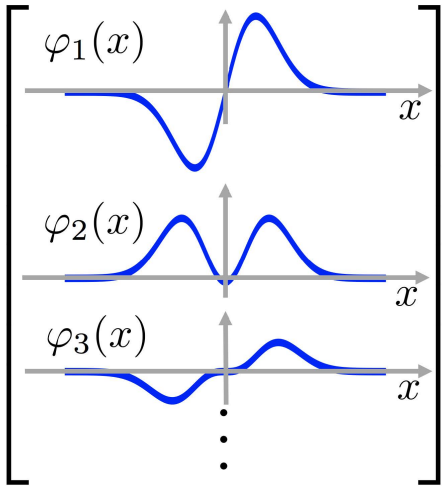

Figure 2: Induced nonlinearities.

Nonlinear Modeling

Nonlinear factor model:

$$\underset{M\times 1}{\varphi(X_t)} = \underset{M\times r}{\Lambda_\varphi}\,\underset{r\times 1}{F_{\varphi,t}} + \underset{M\times 1}{e_{\varphi,t}}, \tag{3}$$

Estimation problem:

$$\begin{gathered} \underset{F_\varphi,\Lambda_\varphi}{\arg\min}\ \left\|\Phi - F_\varphi\Lambda_\varphi'\right\|_F^2 \\ N^{-1}\Lambda_\varphi'\Lambda_\varphi = I_r, \quad F_\varphi' F_\varphi \text{ diagonal.} \end{gathered} \tag{4}$$

However, $\Phi'\Phi$ may not be observable and/or feasible to compute.

Kernel factors:

$\widehat{F}_\varphi = \Phi\Phi' eig_r(\Phi\Phi') = K eig_r(K)$

Kernel factors

Kernel factors:

$\widehat{F}_\varphi = \Phi\Phi' eig_r(\Phi\Phi') = K eig_r(K)$

Compare with Connor and Korajczyk 1993 (nonlinearized):

$$\widetilde{F}_\varphi = \sqrt{T} eig_r(\Phi\Phi'). \tag{5}$$

Both estimators have identical column spaces.

Proposition 1

Estimators $\widehat{F}_\varphi$ and $\widetilde{F}_\varphi$ produce the same projection matrix.

Nesting

Kernel factors nest linear factors.

Proposition 2 (*Simplified*)

$$\lim_{\gamma \to 0} \gamma^{-1} \widehat{F}_{\varphi} W = \widehat{F},$$

for some $W \in \mathbb{R}^{r \times r}$

Connection to NNs

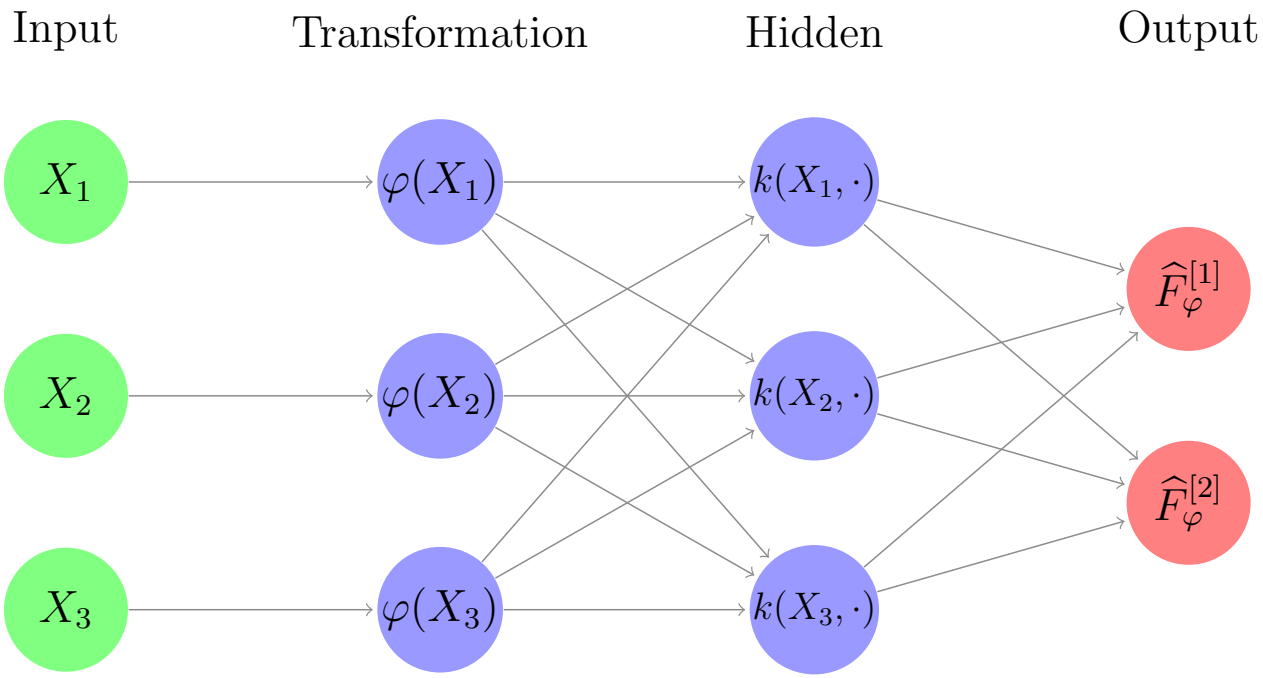

Figure 3: Neural network interpretation of kernel PCA, illustrated for the case $T = 3$, $r = 2$. Each observation is nonlinearly transformed and the inner products are computed. The output units are kernel factors with $\widehat{F}_\varphi^{[j]} = \sum_{i=1}^{T} \alpha_i^{[j]} k(X_i, \cdot)$ which linearly combine these dot products, with the weight estimate calculated as the eigenvector of the kernel matrix K.

Consistency

Two cases:

Case 1: $M < \infty$ (Example, polynomial kernel)

Case 2: $M = \infty$ (Example, RBF kernel)

Consistency: Case 1

Case 1: $M < \infty$

Theorem 1 (Simplified)

Suppose $\varphi(\cdot) : \mathbb{R}^N \to \mathbb{R}^M$, *where* $M := M(N)$ *is such that* $N \leq M(N) < \infty$, *and Assumption A holds. Then as* $M, T \to \infty$

$$\delta_{NT}^2 \left\| \widetilde{F}_{\varphi,t} - H' F_{\varphi,t} \right\|_F^2 = O_p(1)$$

where $\delta_{NT} = \min\{\sqrt{M}, \sqrt{T}\}$.

Consistency: Case 2

Case 2: $M = \infty$

Theorem 2 (Simplified)

Suppose $\varphi(\cdot) : \mathbb{R}^N \to \mathbb{R}^\infty$ and the assumptions of Lemma 1 and 2 hold. Then as $T \to \infty$

$$\sqrt{T} \left\| \widehat{\psi}_i - \psi_i \right\| = O_p(1),$$

where $\widehat{\psi}_i$ and ψ_i are the i-th eigenvectors of $\widehat{\Sigma}$ and Σ respectively

Forecasting Model

$$Y_{t+h} = \beta_0^h + \sum_{p=1}^{P_t^h} \beta_{Y,p}^h Y_{t-p+1} + \sum_{m=1}^{M_t^h} \underset{1\times R_t^h}{\beta_{F,m}^h{}'} \underset{R_t^h\times 1}{F_{t-m+1}} + \epsilon_{t+h}, \tag{6}$$

Three factor equation specifications are considered,

$$\underset{N\times 1}{X_t} = \Lambda F_t + e_t, \tag{7}$$

$$\underset{2N\times 1}{X_{*,t}} = \Lambda_* F_{*,t} + e_{*,t}, \tag{8}$$

$$\underset{M\times 1}{\varphi(X_t)} = \Lambda_\varphi F_{\varphi,t} + e_{\varphi,t}. \tag{9}$$

1. PCA: $\widehat{F}_{PCA} = X eig_r(X'X)$
2. SPCA: $\widehat{F}_{SPCA} = \tilde{X} eig_r(\tilde{X}'\tilde{X}), \quad \tilde{X} = [X, X^2]$
3. kPCA: $\widehat{F}_\varphi = \Phi\Phi' eig_r(\Phi\Phi')$

Hyperparameters: P_t^h, M_t^h, R_t^h and γ for kPCA.

Data

- McCracken and Ng 2016 dataset
- 1959:01 to 2020:04, 736 obs, 128 variables
- All variables made second order stationary
- 8 variables to forecast at $h = 1, 3, 6, 9, 12, 18, 24$
- Rolling windows 120 obs
- Hyperparameters selected by BIC and CV

Kernels:

- Sigmoid $k(\mathrm{x_i}, \mathrm{x_j}) = \tanh(\gamma(\mathrm{x_i}'\mathrm{x_j}) + 1)$
- Radial basis function (RBF) $k(\mathrm{x_i}, \mathrm{x_j}) = e^{-\gamma\|\mathrm{x_i}-\mathrm{x_j}\|^2}$
- Quadratic polynomial (poly(2)) $k(\mathrm{x_i}, \mathrm{x_j}) = (\mathrm{x_i}'\mathrm{x_j} + 1)^d,\ d = 2.$

Empirical Results

Table 1: Relative MSPEs for 8 variables across 7 different prediction horizons. Each value represents the ratio to out-of-sample MSPE of the autoregression augmented by diffusion indexes estimated by PCA.

	$h = 1$	$h = 3$	$h = 6$	$h = 9$	$h = 12$	$h = 18$	$h = 24$
				Real Personal Income			
SPCA	1.0355	1.3427	1.5900	1.7746	1.1645	0.9892	1.2647
PC^2	1.0173	1.1228	1.3011	1.1790	1.2873	1.1590	1.1937
kPCA poly(2)	1.0814	1.9185	2.5423	2.2290	2.1792	1.5982	1.6674
kPCA sigmoid	1.0009	1.0104	0.9823	0.9564	0.9721	0.9895	0.9602
kPCA RBF	0.9993	1.0047	0.9874	0.9941	0.9801	0.9673	0.9774
				Civilian Employment			
SPCA	1.0271	1.4285	1.7018	1.4133	1.7412	2.1467	2.0262
PC^2	0.9918	1.1371	1.2390	1.3073	1.5282	1.3466	1.1950
kPCA poly(2)	1.4346	1.6811	2.8022	2.0038	1.8869	1.6520	1.3763
kPCA sigmoid	1.0065	0.9854	0.9660	0.9600	0.9754	0.9614	0.9065
kPCA RBF	0.9991	0.9804	0.9712	0.9573	0.9711	0.9661	0.9737

Empirical Results

Table 1: Continued

	$h = 1$	$h = 3$	$h = 6$	$h = 9$	$h = 12$	$h = 18$	$h = 24$
Housing Starts: Privately Owned							
SPCA	1.0020	1.4299	1.4810	1.6383	1.1651	1.3553	1.6741
PC^2	1.0698	1.2187	1.5125	1.5515	1.0870	0.9928	0.7737
kPCA poly(2)	1.0634	1.3674	1.6719	1.9320	1.5364	1.8500	2.5680
kPCA sigmoid	0.9996	0.9884	0.9809	0.9916	0.9861	0.8553	0.8356
kPCA RBF	0.9978	0.9838	0.9329	0.8959	0.9014	0.9389	0.9252
Real personal consumption							
SPCA	1.0764	1.3450	1.4787	1.5401	1.4043	1.4061	1.8804
PC^2	1.0467	1.1900	1.3065	1.1856	1.1229	1.1625	1.3006
kPCA poly(2)	1.1722	1.5608	1.7129	1.4684	1.4583	1.0932	1.3171
kPCA sigmoid	1.0021	0.9958	0.9900	0.9729	0.9721	0.9199	0.8921
kPCA RBF	0.9936	0.9942	0.9639	0.9584	0.9750	0.9792	0.9886
M1 Money Stock							
SPCA	1.1004	1.5903	1.3759	1.2666	1.2002	1.0483	1.1778
PC^2	1.1851	1.4353	1.1629	1.1591	2.6222	2.0673	1.5914
kPCA poly(2)	1.9102	2.2750	2.4906	1.9719	1.5915	1.4239	1.4367
kPCA sigmoid	0.9994	0.9785	0.9473	0.9362	0.9577	0.9247	0.9082
kPCA RBF	0.9878	0.9921	0.9563	0.9462	0.9430	0.9556	0.9654

Empirical Results

Table 1: Continued

	$h = 1$	$h = 3$	$h = 6$	$h = 9$	$h = 12$	$h = 18$	$h = 24$
Effective Federal Funds Rates							
SPCA	1.0202	1.4132	2.2041	1.3547	1.6423	2.2325	2.8836
PC^2	1.2822	1.1414	1.4214	1.6993	1.0800	1.0843	1.2805
kPCA poly(2)	1.7056	1.8629	3.4562	3.0524	1.5233	1.8725	1.5638
kPCA sigmoid	1.0072	0.9406	0.8791	0.9430	0.9930	0.9461	0.8629
kPCA RBF	1.0109	0.9696	0.9592	1.0028	0.9943	0.9370	0.8945
CPI: All Items							
SPCA	1.1179	2.0528	2.1043	1.2890	1.1638	1.1911	1.2676
PC^2	1.0175	1.6747	1.1324	1.8631	1.5414	1.0210	1.1420
kPCA poly(2)	0.9850	1.8597	3.9190	1.2954	1.2887	1.2227	1.3105
kPCA sigmoid	0.9909	0.9873	0.9706	0.9826	0.9686	0.9913	0.9839
kPCA RBF	0.9792	1.0166	1.0232	1.0322	0.9788	0.9476	0.9887
S&P's Common Stock Price Index							
SPCA	1.1137	1.3306	1.6338	1.6281	1.5476	1.8519	1.9349
PC^2	1.1150	1.2794	1.4366	1.7406	2.1577	2.0953	1.9181
kPCA poly(2)	1.6275	1.7292	2.4850	2.4820	2.2977	1.9816	1.5089
kPCA sigmoid	0.9842	0.9639	0.9460	0.9268	0.9752	0.9265	0.9637
kPCA RBF	0.9992	0.9888	0.9751	0.9722	0.9677	0.9453	0.9906

Final Remarks

- Constructing factor estimates nonlinearly can be beneficial for macroeconomic forecasting

Possible extensions:

- Targeted variables (Bai and Ng 2008, Bulligan et al. 2015)
- Other dimension reduction techniques
- Dynamic or efficient factors (Forni et al. 2000)

Paper

Contact: varlam@kutateladze.com

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